

MoWa: An Authoring Tool for Refining AI-Generated Human Avatar Motions Through Latent Waveform Manipulation

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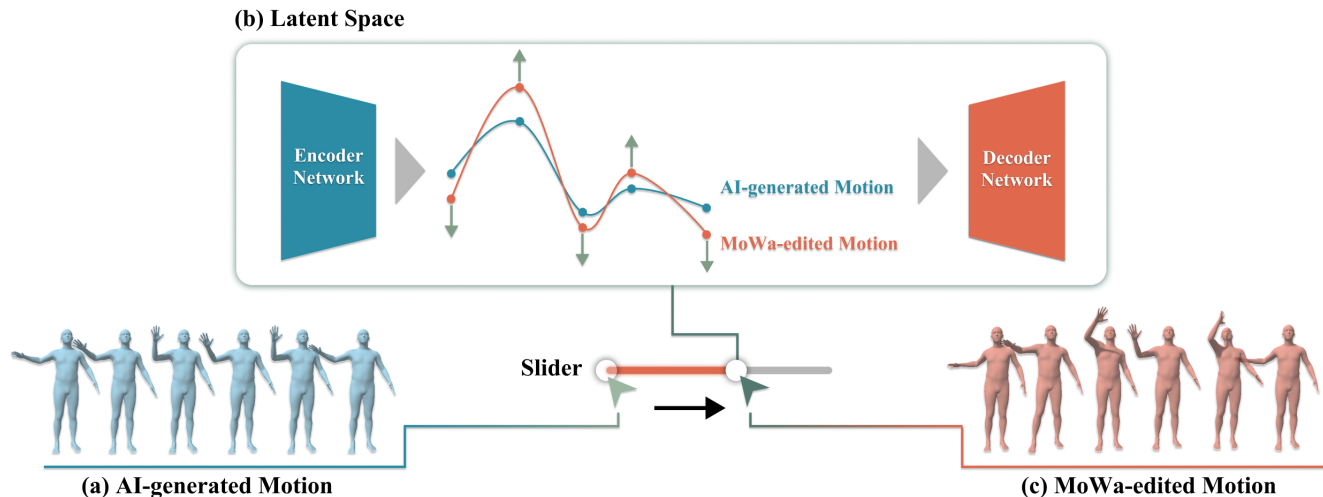


Figure 1: The overall schematic of MoWa: (a) MoWa is a system designed to take AI-generated motions as input and enhance their expressiveness. (b) MoWa maps the motion animations into latent space and modifies the waveform within the latent space. Users can perform this process using a simple UI, such as sliders. (c) Using MoWa, users can adjust the overall expressiveness of the motions based on animation principles.

Abstract

Creating expressive and realistic motion animations is a challenging task. Generative artificial intelligence (AI) models have emerged to address this challenge, offering the capability to synthesize human motion animations from text prompts. However, the effective integration of AI-generated motion into professional designer workflows remains uncertain. This study proposes MoWa, an authoring tool designed to refine AI-generated human motions to meet professional standards. A formative study with six professional motion designers identified the strengths and weaknesses of AI-generated motions. To address these weaknesses, MoWa utilizes latent space to enhance the expressiveness of motions, making them suitable for use in professional workflows. A user study involving twelve professional motion designers was conducted to evaluate MoWa's effectiveness in refining AI-generated motions. The results

indicated that MoWa streamlines the motion design process and improves the quality of the outcomes. These findings suggest that incorporating latent space into motion design tasks can improve efficiency.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

creativity support tool, graphics design, artificial intelligence

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1 Introduction

Creating expressive avatar motion animation is a key task in game and animation development [5, 26]. Expressive motion animations

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on human avatars convey emotions and reactions to given situations [87], improving interaction quality [25, 26], immersion [5, 111], and user engagement [31, 76]. However, producing high-quality human avatar motions demands skilled animators or expensive motion capture technology, both of which require significant effort [52, 113, 115]. To overcome these barriers, text-to-motion techniques have been developed, allowing users to generate human avatar motion animations using artificial intelligence (AI) models [8]. These AI-driven methods make motion creation more accessible and cost-effective, enabling a wider range of creators to produce realistic motions with ease [12].

Text-to-motion techniques enable users to generate human avatar motion animations by inputting descriptive text prompts [41, 92, 114]. These techniques utilize natural language processing to produce motions that are contextually appropriate for the given prompt [34, 35, 64]. The generative models used in this approach are trained on datasets that capture realistic human movements, and the synthesized motions are evaluated based on their similarity to the distribution of these datasets. The reliance on large-scale human motion datasets ensures that the generated motions are grounded in natural human movement [41], making them suitable for applications such as dance motion generation [77], sports motion generation [109], and combat motion generation [71]. Consequently, text-to-motion techniques can streamline the motion design workflow, thereby enhancing both the speed and quality of motion design production.

However, while realism in motion animation is certainly valued, it is not the sole priority in motion design. Instead, finding a balance between realism and expressiveness is considered a major factor [24, 25, 76], as the expressiveness of a motion is important to convey its intentions and influence human preferences [9, 17]. Therefore, expressiveness should be considered when synthesizing motion animations, as human preference plays a key role in evaluating motion quality [85]. To explore this balance, we conducted a formative study in which we presented AI-generated human avatar motions to professional motion animation designers and gathered their feedback. The study revealed that although AI-generated motions were realistic and well-structured, they often missed the expressiveness needed to convey the uniqueness of a motion. Professional designers noted that AI-generated motions frequently missed key animation principles, such as anticipation, exaggeration, follow-through, arcs, and secondary actions. These principles support creating movements that not only appear realistic but also make the animation more convincing [20, 48, 93]. Without these features, AI-generated motions would not meet the expressiveness needed for professional standards.

These findings underscore the need for approaches that bridge the gap between the output of text-to-motion techniques and the expressive requirements of professional motion designers. Therefore, we propose MoWa, an authoring tool designed to enhance the expressiveness of AI-generated motions. To address these limitations of current techniques, which primarily synthesize motions for human avatars, MoWa focuses on refining human motion animations. Figure 1 illustrates how MoWa operates with AI-generated motions. MoWa uses a variational autoencoder (VAE) to leverage latent space for modifying motion animations. Our approach also includes developing an algorithm that enables users to refine AI-generated motions within the latent space and address their weaknesses to

meet professional standards. The main contributions of this study are as follows:

- **Evaluating AI-Generated Motions Based on Professional Designers' Need:** We conducted interviews with professional motion designers to explore how AI can be integrated into their design workflows. Furthermore, we analyze the strengths and weaknesses of current AI-generated motion outcomes based on feedback from our formative study with professional designers.
- **Implementation of MoWa:** We introduce an authoring tool that addresses these weaknesses by incorporating editing algorithms in the latent space. To ensure usability, we provide a user interface for MoWa that allows users to adjust the strength of the algorithm using simple sliders.
- **User Study with Professional Motion Designers:** We conducted a user study with professional motion designers to evaluate MoWa's effectiveness and its impact on the quality and usability of AI-generated human avatar animations. Based on the findings, we discuss implications for further improving and supporting the motion design process by mapping motions to low-dimensional spaces.

2 Related Works

2.1 Creativity Support Tools for Motion Designing

One significant challenge motion designers face is the difficulty in editing human poses [60, 105]. A common approach to human motion design involves creating key poses for the avatar at specific keyframes and interpolating between them. While this method offers flexibility in expressing the designer's intent, it requires extensive manual adjustments [18, 82]. To alleviate this burden, methods have been proposed to edit motion trajectories rather than directly adjusting each joint's rotations [16, 32, 112, 115]. These methods allow designers to modify motion by adjusting the waveform of motion paths, providing a more accessible and intuitive way to create complex motions [16]. Also, systems that incorporate animation principles to simplify the editing process have been proposed. For example, filter-based approaches have been introduced to enhance anticipation and follow-through effects [97, 117], motion editing methods have been developed to incorporate seven animation principles for 2D animations [46], distortion effects have been used to stylize exaggerated effects [10], and algorithms have been designed for stylizing the trajectory animations of objects [63].

Another significant challenge in human avatar motion design is navigating and interacting within a 3D space. Navigating 3D environments using a 2D screen is ineffective due to ambiguity caused by projection [94, 115]. This ambiguity forces designers to frequently interact with the 3D space to target precise positions [55]. To address this issue, studies have proposed methods that utilize extended reality (XR) techniques and 6DoF input devices, enabling more direct and efficient manipulation of 3D spaces [44, 59]. These studies suggest that XR techniques can reduce the complexity of 3D navigation and enhance workflows [115]. However, although these techniques offer promising enhancements to the workflow, there remains a need for effective methods compatible with 2D input

devices, as traditional tools such as screens and mice continue to be prevalent in the industry [68, 101].

In this study, we propose an editing system that utilizes the advantages of trajectory methods and animation principles, integrated with the latent space. While previous researches focused on stylizing motion animations based on individual animation principles, less attention has been given to integrating multiple principles within a single system. To address above challenges, we implemented a system that modifies trajectories in the latent space using multiple animation principles, enabling interactions within the latent space to avoid ambiguity issues associated with direct interactions in the 3D space.

2.2 Creativity Support Tools with Latent Space Exploration

Manipulating latent vectors within the latent space has emerged as a promising approach to generate designs. Network structures such as VAEs, generative adversarial networks, and diffusion models have been developed to generate content from latent vectors, suggesting effective methods to explore the design space [30, 39, 49]. Consequently, utilizing the latent space has emerged as an interaction method for creativity support tools that enables users to synthesize contents. For example, ReelFramer leverages latent spaces to guide journalists in generating scripts and visual storyboards [99]. Chiu et al. introduced an exploration technique for high-dimensional latent spaces by searching in low-dimensional subspaces [15]. SpaceEditing enables users to collaborate with AI models by manually editing the latent space, which leads to better model performance [100].

In addition to these content creation tools, generative models have been widely applied in human motion generation tasks. For example, by combining generative models with language models, text-to-motion techniques can produce motion animations of human avatars aligned with user intent [4, 47, 72]. Supported by large-scale motion capture datasets such as AMASS [64], HumanML3D [34], and KIT-ML [74], these approaches have demonstrated accurate and realistic human motion movements [34, 64] represented in the SMPL format [62]. Recent studies in human motion generation further reveal that incorporating human preferences enhances the quality of text-to-motion outputs [85], highlighting the importance of considering the expressiveness of synthesized human motions.

While manipulating the latent space is effective for synthesizing new content, latent space exploration poses challenges, as it may not be intuitive for users [21, 73]. To address this issue, interfaces that facilitate intuitive exploration without requiring technical expertise should be considered [21, 96, 98]. For interpretable content synthesis, studies have proposed that utilizing a VAE structure is effective, as the latent spaces of these models are efficient in identifying semantic meanings [36, 45, 84]. Consequently, successful creativity support tools have been developed using the advantage of semantic latent spaces [65, 96, 102].

Our study leverages the semantic latent space provided by the VAE structure. Utilizing this semantic latent space, we developed latent space exploration algorithms that integrate animation principles to enhance intuitiveness. Through these algorithms, we aim to improve the expressiveness of motions, making them more convincing and aligned with professional standards.

2.3 Motion Generation Techniques for Expressive Animations

Expressive motion synthesis has been explored through two primary approaches: rule-based algorithms and deep learning-based motion editing. The rule-based algorithmic approach generates new poses by interpolating the angles and positions of joints over time within a motion sequence [7, 29, 48, 80]. This method offers the advantage of interpretability and controllability, as the motion transformations are fully transparent and can be adjusted to meet specific requirements [27].

In contrast, deep learning-based techniques generate expressive motions using datasets. Style transfer methods, for example, allow transferring the style of one pose to another or creating new poses between existing ones [3, 6, 22, 23, 42, 51]. While these methods effectively produce desired styles of motion animations, they depend on high-quality source motions for style extraction, requiring users to capture such motions. Addressing these limitations, the inpainting approach has been introduced, enabling users to edit segments of motion using text prompts [33]. This method allows users to refine motions by providing prompts and further adjusting the generated results [38, 116]. It also facilitates guiding specific body parts of the avatar to perform desired motions via text prompts while preserving the positions and movements of unaffected body parts [33].

Combining these two approaches, we implemented an authoring tool that incorporates rule-based exploration within the latent space of AI-generated motions mapped by an encoder network trained on human motion datasets. Leveraging this approach, we propose an authoring tool that utilizes latent space exploration to refine AI-generated motions.

3 Formative Study

To explore the potential of text-to-motion techniques in the motion design workflow, we conducted a formative study with professional motion animation designers. We asked designers to evaluate human avatar motion animations generated by text-to-motion AI models and provide feedback on how these AI-generated motions could be improved.

3.1 Formative Study: Participants

We recruited six professional designers, each with at least three years of experience in motion animation design. The participants had backgrounds in game development and animation production ranging from 3 to 5 years ($M = 3.83$, $SD = 0.75$). All participants reported having skills in industry-standard software, such as Maya, Cinema4D, or Blender. Detailed information on the participants is provided in Table 1.

3.2 Formative Study: Motion Refining Task

In the formative study, we presented professional motion designers with three different human avatar motions generated by three distinct text-to-motion models [34, 92, 114]. The motions included a punch, a kick, and a walk, each lasting 2 seconds. These motions were selected using prompts commonly showcased by the three models, ensuring that they were representative of each model's capabilities. We designated the primary action for each motion

Table 1: Details about the professional designers who participated in the formative study: Six professional motion designers were recruited to explore the potential for integrating AI-generated motions into professional workflows.

ID	Age	Gender	Experience	Skills
P1	29	Female	4 years	Blender, Cinema 4D, Unreal Engine
P2	25	Male	3 years	Cinema 4D, Marvelous Designer
P3	33	Female	4 years	Blender, Cinema 4D, Unreal Engine
P4	26	Male	4 years	Blender, Unity
P5	34	Female	5 years	Blender, Autodesk Maya
P6	26	Female	3 years	Cinema 4D, ZBrush

Table 2: Details about the motion animations presented in the formative study: Three text-to-motion models were utilized to generate three representative motions.

Item	Prompt for motion	Primary action
Model A (MDM [92])	"A person punches in a manner consistent with martial arts."	Upper-body (Punch)
Model B (T2M [34])	"A person was practicing karate kicks."	Lower-body (Kick)
Model C (MotionDiffuse [114])	"A person is walking slowly forward."	Full-body (Walk)

type as follows: upper-body for the punch, lower-body for the kick, and full-body for the walk. Each motion was generated using prompts identical to those used in the showcase of the respective deep learning models. Table 2 provides the specific prompts and corresponding AI models used to generate the motions presented in the study.

We built a user interface (UI) using the Unity engine to allow professional designers to refine the provided motions by adjusting the joints of an avatar. A detailed view of the implemented UI is shown in Figure 2. We first sampled keyframes from AI-generated motions by identifying the extreme points of the motion curves [103] and displayed them in a timeline panel. By clicking on the keyframe buttons, designers could modify the avatar's pose at each keyframe. Participants were also allowed to add or remove keyframes as needed. In addition, an explanation panel was included at each keyframe, where designers could document their reasons for making specific changes to the motion animations.

To avoid any bias, we did not inform the designers that the given motions were generated by AI. The professional designers were instructed to enhance all three motions until they were satisfied with the results. After completing the refinement process, the designers were then informed that the motions had been generated by AI.

3.3 Formative study: Procedure

The overall process of the formative study is illustrated in Figure 3. Before beginning the task, we asked the designers to provide information on their age, gender, experience, and skills through a pre-questionnaire. After refining each motion, participants completed questionnaires that evaluated the overall quality of the motions. Upon completing all tasks, participants identified specific aspects of their workflow that could benefit from AI assistance through

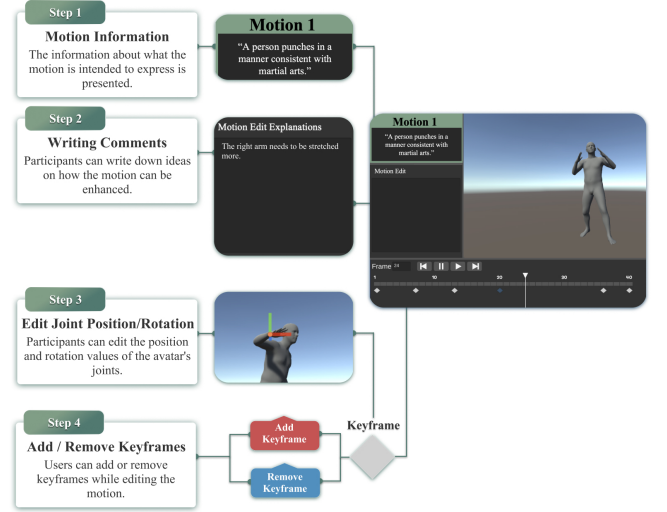


Figure 2: The motion refining task given to the professional designers in the formative study: The designers were instructed to refine the given motions and discuss how the modifications should be made for improvement.

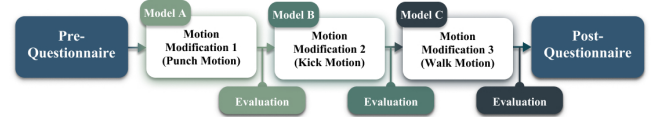


Figure 3: The overall process of the formative study: The professional designers underwent three motion refining tasks to provide insights into how AI-generated motions should be improved.

a post-questionnaire. The questionnaires asked in the formative study are shown in Appendix A–Table 4.

3.4 Formative Study: Findings

3.4.1 Challenges in Motion Designing and Advantages of Integrating AI. Professional designers outlined their workflow, which begins with *posing*, where they create keyframes of the human motion animation. Each keyframe must depict a physically plausible pose, as these poses form the foundation of the animation. After establishing the keyframes, the designers proceed to *splining*, where they interpolate the transitions between the keyframes. This step involves smoothing the movement between poses to ensure that the animation flows seamlessly. Finally, the designers move on to *polishing*, where they review and iteratively refine the motion.

Designers reported that AI could be beneficial to motion design workflow in two main areas. First, all designers highlighted that pose and timing are the most important aspects of the animation. Consequently, creating solid poses with correct timing is considered the most challenging task in motion design: "Creating an appropriate pose is the hardest part of designing (P1, P2, P3, P5)" and "Finding the optimal timing for natural movement is the most challenging task. (P4, P6)". Designers suggested that AI could help bypass the

posing and *splining* stages and move directly to the *polishing* phase, increasing the efficiency of their workflow. Given that AI can accelerate design workflows by automating repetitive tasks involved in the iterative process of refining designs [43, 67, 106], incorporating AI during the *polishing* stage, which involves iterative adjustments and evaluations, can be especially advantageous.

Additionally, designers noted that the process of searching for reference materials is the most time consuming aspect of their work (P1, P3, P6). To create a convincing motion, designers often search for reference videos or images (P1, P3), try out poses themselves (P1, P6), or browse motion libraries such as Mixamo (P6). Designers suggested that AI could enhance their workflow by synthesizing references that can inspire their work. These findings align with previous studies, which have shown that AI-generated outcomes can inspire professionals during the design process [67, 81, 88]. Since the risk of design fixation caused by AI-generated results can also negatively impact designers [19], diversifying AI outcomes is a major factor in supporting professionals' exploration of the design space.

3.4.2 Evaluation of AI-generated Motions. After each motion refinement task, designers were asked to rate the provided motions and discuss their feedback. The results showed that designers rated the punch motion with an average score of 3.50 ($SD = 0.55$), the kick motion with an average score of 2.50 ($SD = 1.05$), and the walk motion with an average score of 2.83 ($SD = 1.17$). Overall, the scores were relatively low, indicating that AI-generated motions require further improvement before they can be fully accepted by professionals. Figure 4 summarizes the strengths and weaknesses of AI-generated motions based on the feedback collected in the formative study.

Among the six professional motion designers, four designers (P2, P3, P4, P6) explicitly reviewed the AI-generated motions by referencing animation principles. They mentioned principles such as anticipation, follow-through, solid pose, slow-in and slow-out, secondary actions, arc, timing, and exaggeration while evaluating the quality of the motions. Based on these observations, we initially categorized the comments of the professional motion designers using a bottom-up approach, using the animation principle keywords as a framework. For the remaining two designers (P1, P5), while they did not explicitly mention the names of the principles, their comments aligned with the concepts of these principles. For example, "*The fist should go further back before delivering the punch*" (P5) aligns with the principle of anticipation. To analyze such comments, we adopted a top-down approach, interpreting and categorizing the designers' feedback based on the categories identified through the bottom-up process.

Strengths of AI-Generated Motions.

- **Solid Pose:** Professional motion designers noted that AI-generated motions produce realistic and physically plausible poses. The AI-generated poses adhere to the anatomical constraints of the human body, allowing designers to bypass the initial stages of manually crafting each keyframe by providing a reliable starting point. This is attributed to text-to-motion AI models being trained on large datasets of human motion capture data, enabling them to replicate the

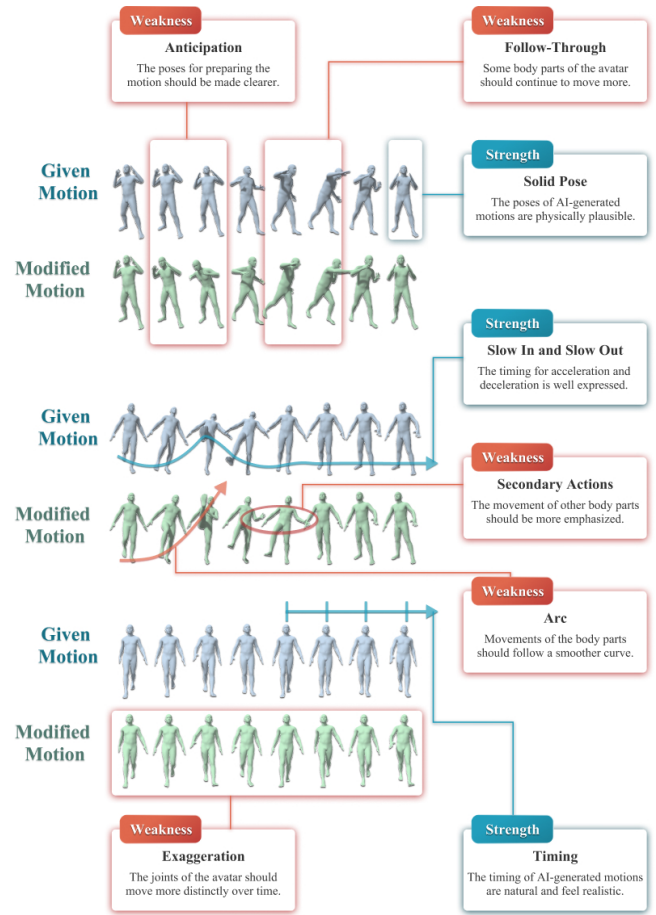


Figure 4: The result of the formative study. Professional motion designers refined the provided AI-generated motions.

physical constraints and natural movement patterns of the human body [11, 69].

- **Timing:** Designers appreciated the timing of AI-generated motions, noting that the intervals between keyframes felt natural, with movements occurring at a pace consistent with real-world physics. For example, in a walking animation, the AI-generated motion accurately captured the rhythm of footfalls and arm swings, resulting in realistic movements.
- **Slow-in and Slow-out:** AI-generated motions effectively incorporate the principles of gradually accelerating into a motion and decelerating out, which contributes to a lifelike animation. This smooth interpolation is achieved because AI models are optimized for temporal consistency using specialized architectures or loss functions designed to handle temporal data [25, 90].

Weaknesses of AI-Generated Motions and Suggestions for Enhancement.

- **Arc:** Arcs refer to the smooth, curved trajectories that natural movements follow. Designers observed that AI-generated

motions sometimes failed to maintain smooth arcs, particularly in actions such as punching and kicking. For example, the trajectory of a kick might include unnecessary deviations, disrupting the smoothness of the motion.

- **Exaggeration:** Exaggeration amplifies actions or poses to make them more dynamic and visually engaging. While AI-generated motions are often realistic and natural, designers noted that they sometimes fall short in providing the exaggerated movements. For instance, in a walking motion, designers suggested that making the strides longer or the arm swings more pronounced could improve the animation.
- **Anticipation:** Anticipation involves preparatory actions that occur before a major movement. Designers observed that while AI-generated motions include elements of anticipation, these are often under-emphasized and could be more pronounced. For example, in a punching motion, the AI-generated animation might show a slight preparatory movement before the punch, but this anticipation is often subtle, diminishing the motion's impact and engagement.
- **Follow-Through:** Follow-through refers to the continuation of a motion after the main action is completed. Designers noted that while AI-generated motions often include follow-through, it is typically not pronounced enough, especially in actions like punching or kicking. For example, in a punching motion, the arm movement does not fully express the momentum after the punch, causing the motion to stop more abruptly than it should.
- **Secondary Actions:** Secondary actions are smaller movements that occur alongside the primary action. Designers pointed out that AI-generated motions often miss sufficient secondary actions. For instance, in a punching motion, while the arm may move correctly, the rest of the body, such as the legs and torso, remain static, diminishing the overall expressiveness of the motion.

4 MoWa

4.1 System overview

Figure 5 shows the overall user interface of MoWa. Users begin by generating desired human avatar motion animations using three different text-to-motion models [34, 92, 114] (Figure 5 – (a)), each of which processes the prompt to produce a unique motion sequence. Through MoWa, users can adjust motion features of AI-generated motions such as anticipation, follow-through, arcs, exaggeration, and secondary motions (Figure 5 – (b) Waveform Manipulation). The system offers a set of sliders based on animation principles identified during the formative study, enabling users to adjust the overall feel of the motion through slider configurations. For more granular control, the system also supports editing at the keyframe level, which enables users to fine-tune individual poses within the motion (Figure 5 – (b) Fine Adjustment). MoWa provides a fine adjustment UI in the form of a 2D gradient rectangle for each keyframe, allowing detailed modifications of latent vectors at each frames.

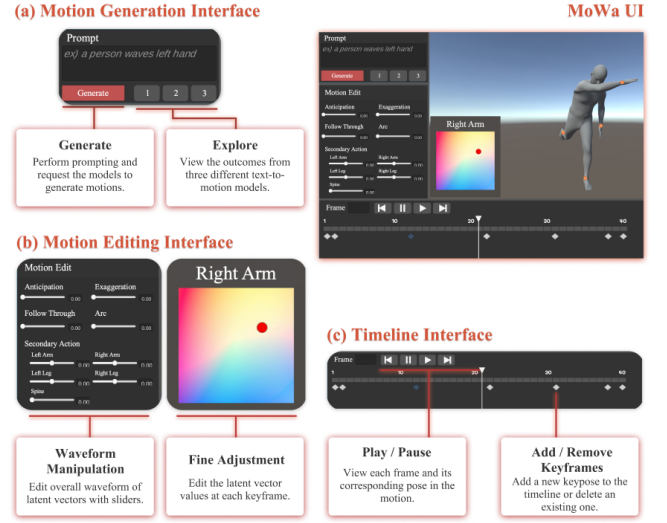


Figure 5: The MoWa user interface: (a) Users can explore the motion design space using text-to-motion models, (b) For AI-generated motions, MoWa allows adjustments based on animation principles with the motion edit interface, (c) Users can freely play, pause, or add keyframes to the motion using the timeline interface.

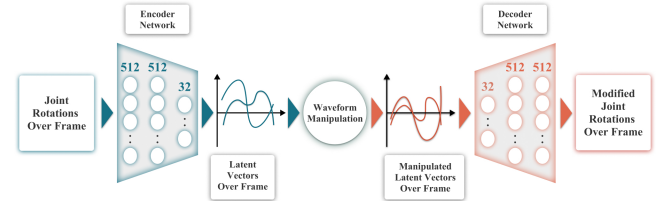


Figure 6: The structure of MoWa: MoWa first uses a VAE to map the 3D space motion animation into a latent space, where waveform modifications are performed. The modified latent vector is then mapped back to the 3D world space to produce the edited motion.

4.2 Projecting a Motion into Latent Space

The AI-generated human motion animations are represented as a time series of SMPL's 21 joints, each described by quaternion values. For a motion animation with N frames, this representation results in an $N \times 84$ dimensional time series data, where 84 corresponds to the 21 joints multiplied by the 4 dimensions of a quaternion. MoWa utilizes a VAE network illustrated in Figure 6, which maps each human pose into a latent vector. The VAE's encoder consists of two 512-dimensional linear layers followed by a 32-dimensional linear layer, compressing the input data into the latent space. The decoder expands the 32-dimensional latent vector back through reversed layers to reconstruct the original data. The model was trained with the AMASS dataset, a superset of human motion datasets that integrates and standardizes motion capture datasets. We trained the model using an RTX 3090 environment, and the training process

optimized the following loss function:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q(z|x)} [\|x - \hat{x}\|^2] + \mathbf{D}_{\text{KL}}(q(z|x) \parallel p(z)) \quad (1)$$

where x denotes the original SMPL pose represented as an 84-dimensional quaternion vector, and $\hat{x}$ is the reconstructed pose generated by the decoder. The first term represents the reconstruction loss measuring the difference between the original and reconstructed poses, and the second term is the KL divergence ensuring the latent space distribution remains close to the prior $p(z)$, typically a Gaussian distribution.

Each frame's pose, originally expressed as an 84-dimensional quaternion vector, is compressed into a 32-dimensional latent vector. The 32-dimensional latent space was chosen to balance dimensionality reduction and reconstruction performance. Latent spaces for VAEs are often mapped to dimensions such as 32 or 64, which are powers of 2, for computational efficiency [91, 95]. We selected 32 dimensions to compress the original data while avoiding an overly low-dimensional mapping that could degrade the quality of the reconstructed motion. In the encoding process, the motion animation was not compressed along the temporal axis, making each element of the latent vector to form a waveform over time. The primary goal of our system is to enable users to refine AI-generated motions frame by frame while preserving the overall structure of the motion and incorporating adjustments based on animation principles. Mapping world space motion animation into latent space offers several advantages for editing motion animations:

- **Dimensionality Reduction:** The world space joint rotation vectors are projected into a latent vector with lower dimensionality, simplifying manipulation and capturing the underlying structure of the motion data [86]. Combined with Jacobian analysis discussed in Section 4.3, this approach provides an efficient representation of motions, allowing the system to generate animations without the need for high-dimensional data.
- **Physical Plausibility:** VAE inherently learns the physical constraints of human anatomy by training on large motion capture datasets [64, 70, 78, 109]. A trained VAE can act as a prior for human motion, ensuring that any modifications made within the latent space adhere to natural anatomical constraints. Consequently, the VAE approach guarantees that physically plausible poses are synthesized when refining a motion.

4.3 Motion Analysis

Jacobian Analysis. While the VAE reduces the dimensionality of motion representation by mapping it to a latent space, the resulting dimensionality is still relatively high, requiring further investigation to understand its structure and influence. To address this, we analyzed the relationship between the latent vector $z(t)$ at a given frame t and the decoded pose. Additionally, to understand the direct influence of the latent space on the pose and enable body part-specific analysis in the algorithm introduced later, we performed Jacobian analysis to identify which elements of the latent vector most impact each joint. Given a motion animation sequence with T frames, the encoder maps the pose vector at the t -th frame to the latent vector $z(t)$, where $0 \leq t \leq T$. For the latent vector

$z(t)$, we computed the Jacobian matrix $\mathbf{J}(t)$ of the decoder with respect to $z(t)$. To evaluate the overall influence of each element of the latent vector on the pose vector elements, we averaged the Jacobians across all frames.

$$\bar{\mathbf{J}} = \frac{1}{T} \sum_{t=0}^T \mathbf{J}(t), \quad \text{where } \mathbf{J}(t) = \frac{\partial g}{\partial \mathbf{z}(t)}, \quad (2)$$

The i -th row of the Jacobian, denoted as $\bar{\mathbf{J}}_i$, represents the influence of the 32-dimensional latent vector elements on the i -th element of the pose vector in the world coordinate. For each body part of the SMPL avatar, we identified the top n indices among $\bar{\mathbf{J}}_i$ with the highest influence on the i -th element of the pose vector by selecting the indices corresponding to the n -largest absolute values in $\bar{\mathbf{J}}_i$:

$$\mathbf{Z}_i = \{j_1, j_2, \dots, j_n\} \quad \text{such that} \quad |\bar{\mathbf{J}}_{i,j_k}| \geq |\bar{\mathbf{J}}_{i,j}| \quad \text{for } j \notin \mathbf{Z}_i, \quad (3)$$

where $\mathbf{Z}_i$ denotes the set of the top n indices of the latent vector for the i -th pose element. In our implementation, we set $n = 5$. We observed that in our trained VAE, at most 5 latent vector elements significantly contribute to each joint, and for the sake of simplicity, we uniformly restrict n to 5 for all joints. This process reduces the dimensionality of the latent vector for each joint to n , enabling more efficient exploration within the latent space.

Primary Action Identification. The VAE normalizes elements of the latent vectors $z(t) = [z_1(t), z_2(t), \dots, z_{32}(t)]$ to have a mean of 0 and a standard deviation of 1. This ensures a fair comparison between body parts with different degrees of freedom and amplitudes of natural movement. For example, comparing the wrist joint to larger joints like the spine in terms of rotation in the world coordinate system may be impractical, as the wrist joint typically has a much greater range of motion. To identify the body part involved in the motion's primary action, the fluctuation of each body part's latent vector components is calculated over time:

$$f_i = \sqrt{\frac{1}{T} \sum_{t=1}^T (z_j(t) - \mu_j)^2}, \quad j \in \mathbf{Z}_i, \quad (4)$$

where f_i represents the fluctuation of the latent vectors for the i -th body part, $z_j(t)$ denotes the latent vector component for the i -th body part at time t , μ_j is the mean value of the latent vector component over time, and T is the total number of frames. By comparing the fluctuations, our algorithm analyzes joints related to the primary action.

4.4 Latent Space Exploration Algorithms

Once the motion has been generated and mapped into the latent space, MoWa offers a range of algorithms to explore the latent space and refine the animation, which are shown in Figure 7. Since the latent space is less intuitive for users compared to the world space, we associated latent space exploration with animation principles to enhance user understanding. In this setup, a single motion mapped into the 32-dimensional latent space can be represented as a $32 \times \text{frame number}$ time series. Each element of the latent vector forms a waveform over time. We analyzed the extrema of

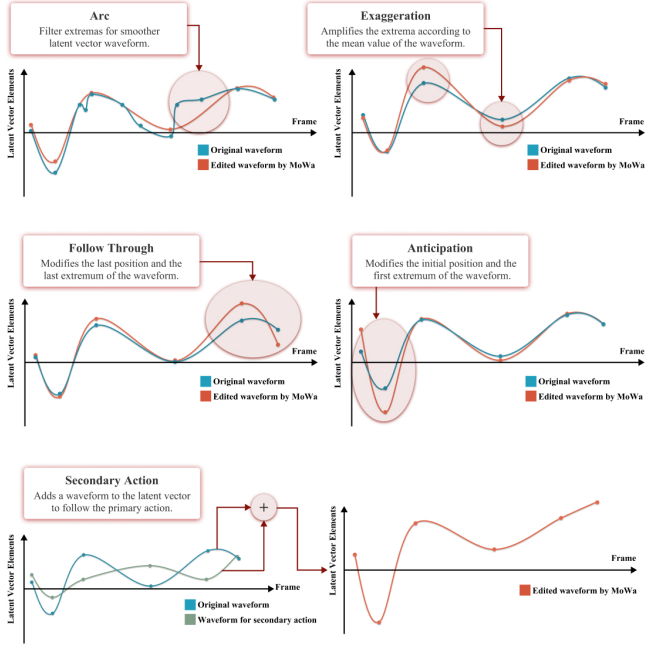


Figure 7: A schematic of waveform manipulation: MoWa edits the waveform of the latent vector according to the animation principles.

each waveform and designed a set of algorithms that allow users to adjust these extrema to reshape the motion accordingly. These algorithms are tied to animation principles, supporting designers easily interpret and refine the resulting motion. To support comprehensive motion refinement, MoWa provides **waveform manipulation** algorithms, which enable users to modify multiple frames simultaneously, altering the general feel of the motion. For more detailed refinement, MoWa offers a **fine adjustment** algorithm to latent vectors, allowing precise modifications to the waveforms of latent vectors in the latent space. Example results of these algorithms are shown in Figure 8.

4.4.1 Waveform Manipulation. Analyzing the Jacobians helps reduce the dimensionality of the latent space for exploration. However, because the VAE does not compress motion animations along the temporal axis, users may encounter difficulties when navigating the latent space across all frames. To address this issue, MoWa provides algorithms that enable users to manipulate the waveforms of latent vectors over time by adjusting a set of scaling factors, offering control over temporal modifications in the latent space. The waveform manipulation algorithms are designed to align with the animation principles identified as underrepresented during our formative study with professional designers, thereby enhancing the intuitiveness of latent space exploration.

Below, the algorithms operate on latent vectors over time, $\mathbf{z}(t) = [z_1(t), z_2(t), \dots, z_{32}(t)]$ for i -th component of the pose vector in the world space.

- **Arc:** The waveform of the latent vector over time sometimes contain noises that disrupt the smooth trajectory of

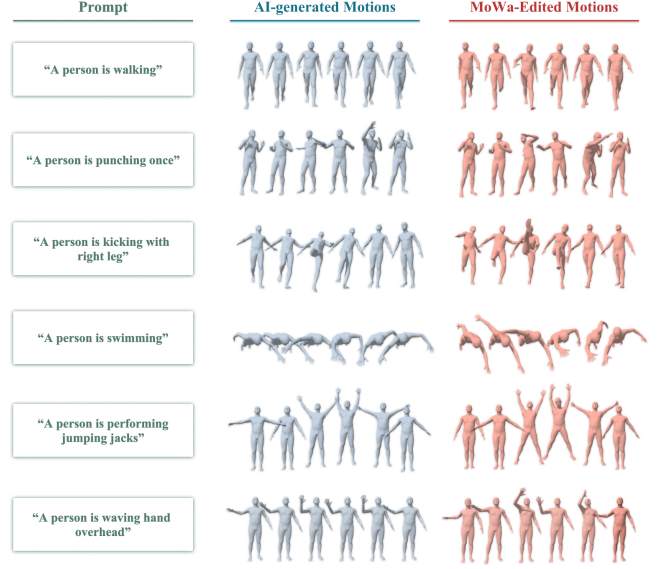


Figure 8: Example results of MoWa: MoWa enhances the expressiveness of AI-generated motions synthesized from given prompts through latent waveform manipulation algorithms.

the motion in world space. To address this, MoWa applies a smoothing algorithm to the latent vector over the entire range of motion. This smoothing is achieved using a sliding window average, represented by the following equation:

$$\mathbf{z}'_j(t) = \frac{1}{2C_{z_j}^{Arc} + 1} \sum_{t'=-C_{z_j}^{Arc}}^{C_{z_j}^{Arc}} \mathbf{z}_j(t+t') \quad \text{for } j \in \mathbf{Z}_i, \quad 0 \leq t \leq T \quad (5)$$

where $\mathbf{z}'_j(t)$ represents the adjusted latent vector, $C_{z_j}^{Arc}$ represents the window size, which controls the extent of the smoothing effect and T is the length of the animation sequence. When exploring the latent space of the trained VAE continuously, the decoded avatar poses in the world coordinate system exhibit corresponding continuous changes [13, 70]. Therefore, removing noise from the waveform in the latent space results in motion trajectories in the world coordinate system that align with the modified latent space, producing smooth motions.

- **Exaggeration:** MoWa applies exaggeration to motion animations by amplifying all extrema within the motion sequence, excluding the first and last extrema. The adjustment for exaggeration can be mathematically described as follows:

$$\mathbf{z}'_j(t_{Ext}) = \mathbf{z}_j(t_{Ext}) + C_{z_j}^{Exaggeration} \cdot (\mathbf{z}_j(t_{Ext}) - \mu(\mathbf{z}_j(t))) \quad \text{for } j \in \mathbf{Z}_i, \quad t_{FirstExt} < t < t_{LastExt} \quad (6)$$

where $\mathbf{z}'_j(t)$ represents the adjusted latent vector, $\mu(\mathbf{z}_j)$ represents the mean value of the latent vector over time, $C_{z_j}^{Exaggeration}$

is a scaling factor for exaggeration, $t_{FirstExt}$ is the frame of the first extremum and $t_{LastExt}$ is the frame of the last extremum. Previous studies have implemented exaggerated motions by creating extrapolated motion trajectories. These extrapolated motions amplify the trajectory, making the intent of the motion appear more pronounced and clear [53, 54]. Similarly, our exaggeration algorithm enhances the trajectory within the latent space, ensuring that the decoded motion in the world space features larger and more expressive trajectories.

- **Anticipation:** To improve anticipation, MoWa identifies extrema in the latent vectors, which represent key moments where the motion’s direction or intensity changes. The anticipation algorithm amplifies the difference between the initial latent vector value and the first extremum. This adjustment enhances the sense of buildup and readiness in the animation. The mathematical formulation for anticipation is as follows:

$$\mathbf{z}'_j(t) = \mathbf{z}_j(t) + C_{z_j}^{Anticipation} \cdot (\mathbf{z}_j(t_{FirstExt}) - \mathbf{z}_j(0)) \quad \text{for } j \in \mathbf{Z}_i, \quad 0 \leq t \leq t_{FirstExt} \quad (7)$$

where $\mathbf{z}'_j(t)$ represents the adjusted latent vector, $t_{FirstExt}$ is the frame of the first extremum, and $C_{z_j}^{Anticipation}$ is a scaling factor for anticipation. This algorithm is inspired by previous work that introduced additional curvature at the beginning of joint angle variations over time to create anticipation in motion [48]. For AI-generated motions, anticipation is often present but may require amplification in intensity. In our approach, instead of introducing additional extrema by creating new splines, we focused on enhancing the existing splines to strengthen the anticipation effect.

- **Follow-Through:** The follow-through algorithm amplifies the difference between the final value of the latent vector and the last extremum. This adjustment enhances the natural continuation of motion beyond the primary action. The mathematical formulation for follow-through is as follows:

$$\mathbf{z}'_j(t) = \mathbf{z}_j(t) + C_{z_j}^{Follow-Through} \cdot (\mathbf{z}(T) - \mathbf{z}(t_{LastExt})) \quad \text{for } j \in \mathbf{Z}_i, \quad t_{LastExt} \leq t \leq T \quad (8)$$

where $\mathbf{z}_j(T)$ is the final value of the latent vector, $t_{LastExt}$ is the frame of the last extremum, T is the frame length of the animation sequence, and $C_{z_j}^{Follow-Through}$ is a scaling factor for the intensity of the follow-through effect. Our follow-through algorithm was inspired by a previous work [97] that proposed an algorithm to create squash and stretch effects. While the prior study focused on squash and stretch, we observed that the approach of scaling the final portion of the world space motion waveform to produce elastic movements could also be applied to follow-through effects. Based on this insight, we developed an algorithm that enhances follow-through by adapting the principles of scaling motion waveforms inside the latent space.

- **Secondary Action:** We designed the secondary action algorithm to be generated based on the primary action. This

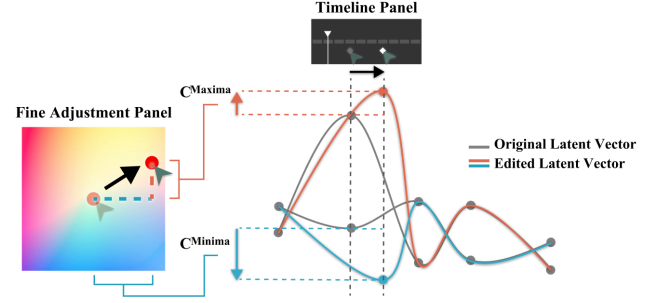


Figure 9: The UI of fine adjustment algorithm. Users can move the position or modify the value of the extremum point of the latent vector at each keyframe.

approach aligns with the design principle that secondary actions should be subordinate to the primary action [57]. Therefore, MoWa analyzes the rotation axis of the body part performing the primary action. For each frame t in the animation sequence $0 \leq t < T$, MoWa calculates the rotation axis between the positions of the primary action’s joints at frame t and $t + 1$. Our algorithm computes the local position of each avatar joint relative to the pelvis. Given the local positions of the joints for the body part performing the primary action at time t , $\mathbf{p}_t$, and at time $t + 1$, $\mathbf{p}_{t+1}$, we calculate the rotation angle:

$$\theta^{Primary}(t) = \arccos \left(\frac{\mathbf{p}_t \cdot \mathbf{p}_{t+1}}{\|\mathbf{p}_t\| \|\mathbf{p}_{t+1}\|} \right), \quad (9)$$

where $\theta^{Primary}(t)$ is the rotation angle of the primary action. Once the rotation of the primary body part is identified, the system applies a similar motion pattern to adjacent joints to boost the overall expressiveness of the animation. This adjustment can be represented mathematically as follows:

$$\theta'_i(t) = g(f(\theta_i(t) + C^{Secondary} \cdot \theta^{Primary}(t))) \quad (10)$$

where f is the encoder network, g is the decoder network, $\theta_i(t)$ represents the rotation of the secondary joints, and $C^{Secondary}$ is a scaling factor that controls the influence of the primary motion on the secondary action. After applying this motion pattern to the other body parts, the resulting rotation vector is processed through the VAE to ensure that the motion remains within plausible poses.

Waveform Fine Adjustment at Each Keyframe. While waveform manipulation simplifies the exploration process simultaneously editing multiple latent vectors, it also introduces a limitation where users cannot explore the entire latent space. To overcome this limitation, we implemented fine adjustment algorithm and a user interface that support direct modifications to the time series of the latent vector. For each keyframe, our fine adjustment algorithm analyzes whether the latent vector associated with that frame belongs to a local minima or maxima. The algorithm then amplifies the values in each minima and maxima group by a constant factor relative to their mean values which can be represented as follows:

$$\mathbf{z}'_j(t) = \begin{cases} \mathbf{z}_j(t) + C_{z_j}^{minima} \cdot (\mu(\mathbf{z}_j(t)) - \mathbf{z}_j(t)), & \text{if } j \in \mathbf{Z}_i, \\ & \mathbf{z}_j(t) \in \text{minima}, \\ \mathbf{z}_j(t) + C_{z_j}^{maxima} \cdot (\mathbf{z}_j(t) - \mu(\mathbf{z}_j(t))), & \text{if } j \in \mathbf{Z}_i, \\ & \mathbf{z}_j(t) \in \text{maxima}. \end{cases} \quad (11)$$

where C^{minima} and C^{maxima} are the scaling factors for fine adjustment. Figure 9 shows the UI of the waveform fine adjustment algorithm. The rectangular 2D panel lets users manipulate the position of a point to control the C^{minima} and C^{maxima} values during latent space exploration. The x coordinate of the point adjusts C^{minima} , while the y coordinate adjusts C^{maxima} , allowing users to fine-tune the poses at each keyframe according to their preferences. The timeline interface allows the users to edit the position of the extremum alongside the temporal axis.

Determining the Scaling Factors from User Input. From the Jacobian analysis, the average Jacobian of the decoder network $\bar{\mathbf{J}}$ was calculated. The i -th row of $\bar{\mathbf{J}}$ indicates the impact of each latent vector element on the i -th element of the pose vector. This information was utilized to determine the scaling factors for the animation principle algorithms. Our algorithm first normalizes the rows of the Jacobian matrix using min-max normalization as follows:

$$\bar{\mathbf{J}}_i^{norm} = \frac{\bar{\mathbf{J}}_i - \min(\bar{\mathbf{J}}_i)}{\max(\bar{\mathbf{J}}_i) - \min(\bar{\mathbf{J}}_i)}, \quad (12)$$

where vector $\bar{\mathbf{J}}_i^{norm}$ is the normalized i -th row of the Jacobian where all elements lie between 0 and 1. The user-defined values set through MoWa's sliders are applied to adjust the latent vectors. For each i -th element of the pose vector, the normalized Jacobian serves as a weight to scale the waveform changes based on the user's slider input α :

$$C_{z_j} = \begin{cases} \text{round}(10 \cdot \alpha), & \text{if } C_{z_j} \text{ is } C_{z_j}^{Arc}, \\ \alpha \cdot \bar{\mathbf{J}}_{i,j}^{norm}, & \text{otherwise.} \end{cases} \quad (13)$$

where $\bar{\mathbf{J}}_{i,j}^{norm}$ represents the j -th element of $\bar{\mathbf{J}}_i^{norm}$. α is a scaling factor that can take values between 0 and 1 for C^{Arc} , $C^{Exaggeration}$, $C^{Anticipation}$, and $C^{Follow-Through}$, while for $C^{Secondary}$ and fine adjustment, α can range between -1 and 1. When text-to-motion models synthesize human motion animations, we sample α to modify the AI-generated motions, allowing the system to present users with more diverse results:

$$\alpha = \begin{cases} 2, & \text{if } C \text{ is } C^{Arc}, \\ \mathcal{N}(0, 1), & \text{if } C \text{ is } C^{Secondary}, C^{minima}, \\ & C^{maxima}, \\ |\mathcal{N}(0, 1)|, & \text{if } C \text{ is } C^{Exaggeration}, C^{Anticipation}, \\ & \text{or } C^{Follow-Through}. \end{cases} \quad (14)$$

where $\mathcal{N}(0, 1)$ denotes a Gaussian distribution with mean 0 and standard deviation 1.

Interpolation Functions. To refine the motion, our algorithm employs interpolation functions that modify the latent waveforms while preserving the overall integrity of the original data. In particular, we apply a vertical transformation to adjust the amplitude of latent vector segments, a horizontal transformation to alter the duration of these segments, and a smoothing operation at segment boundaries to ensure seamless transitions. Detailed formulations of these interpolation functions are provided in the Appendix B.

5 User Study

The user study was designed to evaluate the effectiveness of MoWa through two distinct tasks: Task 1 (T1) motion generation task, and Task 2 (T2) motion editing task. T1 was designed to assess how well MoWa enhances the expressiveness of AI-generated motions. Participants were asked to generate human avatar motions using our system and evaluate the improvements made by the motion correction algorithm. T2 focused on analyzing the effectiveness of editing capabilities of MoWa by allowing participants to refine AI-generated motions.

5.1 Participants

We recruited 12 professional motion designers, including five males and seven females. None of these participants were involved in the formative study, and vice versa. The average age of the participants was 27.92 ($SD = 3.63$), with ages ranging from 24 to 35. The average professional experience was 3.75 years ($SD = 1.48$), with experience ranging from 2 to 7 years. For the user study, we set up a server equipped with an RTX 3090 GPU, capable of running the text-to-motion models used in our system. Participants accessed the client application via a web browser, which allowed them to remotely connect to the server and generate motions using our system.

5.2 T1: Motion Generation Task

5.2.1 T1: Task. The goal of T1 was to compare the unmodified outcomes of the AI-generated motions (**Baseline**) with the outcomes that had been automatically refined by our system (**MoWa**). Participants were provided with simple descriptions explaining the motions they needed to generate with text-to-motion models outlined in Table 3. Participants continued prompting the models until they found a motion that was either satisfactory or suitable for further modification. The keywords for the motions were selected from common showcase examples provided by the text-to-motion models: *throwing* to validate upper-body movement, *jumping* to validate lower-body movement, and *running* to validate full-body movement using MoWa.

For each motion, participants completed the generation process twice: once under the *Baseline* condition, where the outcome was directly obtained from the AI model without any modifications, and once under the *MoWa* condition, where the outcome was automatically refined by our system. The order of these conditions was counterbalanced to avoid learning effects. Additionally, participants were unaware of which condition they were working under during the prompting process.

5.2.2 T1: Measures. After generating a motion under each *Baseline* and *MoWa* condition, participants evaluated the outcomes using the questionnaire shown in Table 5 – Evaluation. The questionnaire

Table 3: Motions to be generated in T1 using Baseline and MoWa: The motion types were selected from those commonly demonstrated by text-to-motion models.

Motion	Provided target motion description	Primary action
Running	Create a motion of a character running forward.	Full-body
Jumping	Create a motion of a character jumping up and down.	Lower-body
Throwing	Create a motion of a character throwing something with right hand.	Upper-body

included questions about satisfaction with the outcome, the quality of the results and workflow, and questions inspired by the creativity support index (CSI) [14] focusing on the exploration experience. Additionally, we recorded the number of prompts and the overall exploration completion time. These measures were collected to assess the efficiency of exploring the motion results provided by the text-to-motion models. Following the design tasks under both *Baseline* and *MoWa* conditions, participants completed a comparison questionnaire, as shown in Table 5 – Comparison.

5.2.3 T1: Results. Since the Shapiro-Wilk test indicated that the data were non-parametric, we conducted a Wilcoxon signed-rank test to assess the significance between the *Baseline* and *MoWa* conditions. Significance levels are indicated as $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Detailed statistics and test results are presented in Appendix C– Table 7.

Fewer prompt attempts, faster exploration. As presented in Figure 10 (a) and (b), participants achieved satisfactory motions with significantly fewer prompt attempts under the *MoWa* condition ($p = 0.029^*$), leading to faster task completion compared to the *Baseline* condition ($p = 0.012^*$). This reduction in prompt attempts was consistent across all specific motion tasks: running ($p = 0.040^*$), jumping ($p = 0.034^*$), and throwing ($p = 0.022^*$). Similarly, exploration time was significantly reduced in the *MoWa* condition for jumping ($p = 0.042^*$) and throwing ($p = 0.016^*$). These efficiency improvements were reflected in the exploration metric (Figure 10 (c) - Exploration), where participants reported greater satisfaction in exploring with the *MoWa* condition ($p < 0.001^{***}$), indicating an improved user experience in exploration.

Higher satisfaction toward the outcome and the system. As shown in Figure 10 (c), outcome satisfaction was significantly higher in the *MoWa* condition ($p = 0.007^{**}$), reflecting that the motions generated were more aligned with the participants' expectations. Similarly, outcome diversity, which measures the variety of motions produced, was also rated higher in the *MoWa* condition ($p = 0.001^{**}$), indicating that participants found the system capable of generating a broader range of acceptable motions. Outcome quality, assessing the overall quality of the motions, showed a significant improvement in the *MoWa* condition ($p = 0.007^{**}$), and higher scores are reported in intention reflection ($p = 0.002^{**}$). Participants expressed a stronger preference of using the system in the future ($p = 0.008^{**}$). These results suggest that *MoWa* could provide outcomes that were more satisfying and aligned with the designers' creative intentions.

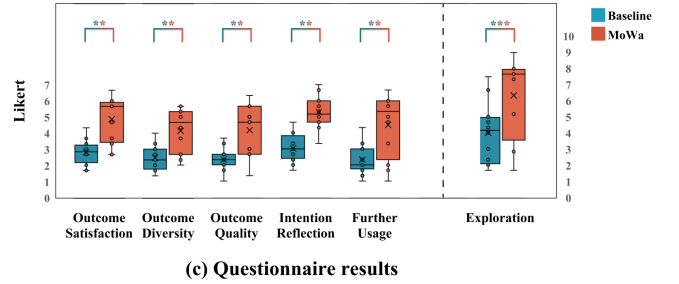
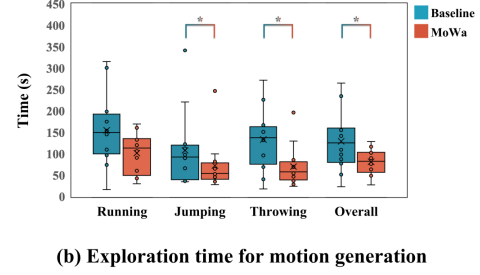
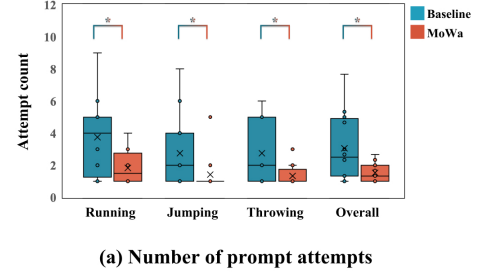


Figure 10: Results from T1: (a) Number of operations performed by the designers using *Baseline* and *MoWa*. (b) Time taken for designers to find a motion they were satisfied with using *Baseline* and *MoWa*. (c) Evaluation questionnaire results reported by the designers. Significance levels are indicated as follows: $*p < 0.05$, $p < 0.01$, $***p < 0.001$.**

Higher preference toward MoWa outcome. The comparison questionnaire revealed a clear preference for the *MoWa* condition over the *Baseline* condition as shown in Figure 11. When asked about reasons for their preference, participants who favored *MoWa* highlighted that the poses were more dynamic and easier to recognize. The participants noted that while the *Baseline* motions were natural, the motions felt constrained and missed the movement needed to clearly convey the intended actions: "The motion doesn't have enough movement to properly express what I intended" (P2, P4, P7), "The running motion misses exaggeration, making it hard to clearly see the running posture." (P5), and "The jumping motion should involve a deeper crouch before leaping off the ground." (P4). These designers have also reported that it would be easier to refine the *MoWa* motion compared to the *Baseline*, and the result better matched their prompt intentions: "It seems to align better with what I asked for in the prompt" (P2, P7, P8), "I like the expression of force in

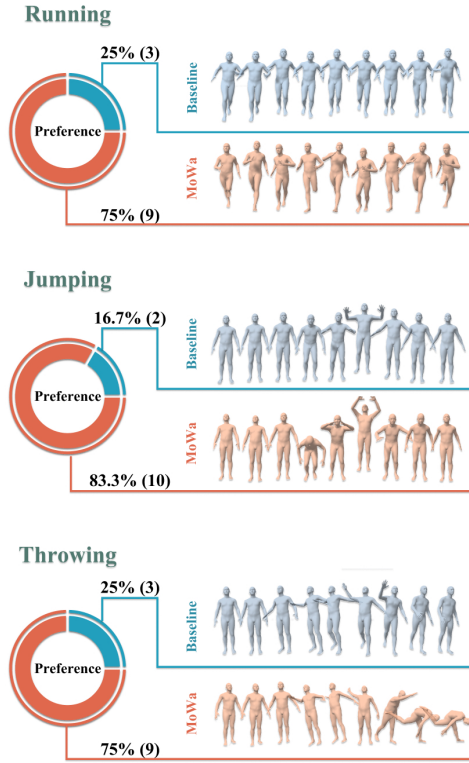


Figure 11: The results of the comparison questionnaire from T1: A higher percentage of participants preferred the outcome generated by MoWa compared to that of Baseline.

the throwing motion" (P5), and "The timing and the speed expression in the motion is appealing" (P9, P10). In addition, some participants found MoWa motions to be more realistic than those of the Baseline: "The motion feels more realistic, with more appropriate expressions" (P4, P9).

However, other participants preferred the Baseline over MoWa. These participants appreciated the naturalness of the Baseline motions, noting that MoWa's results were overly exaggerated: "The motion is too exaggerated, which actually lowers the overall quality" (P6, P11) and "I prefer the realism of the Baseline motion" (P11). These findings suggest that while MoWa was generally favored for its dynamic and expressive qualities, the Baseline's realistic and subtle approach also held appeal, highlighting the importance of balancing expressiveness with realism in motion design.

5.3 T2: Motion Editing Task

5.3.1 T2: Task. In T2, participants were asked to take the throwing motion they selected in T1 and modify it to create a more desirable motion using three different editing systems: *Inpainting*, *Manual*, and *MoWa*. Each system was implemented using the Unity engine and distributed as a client application that communicated with our server. Throughout the task, participants were asked to refine the throwing motion they selected in T1 until they were satisfied with

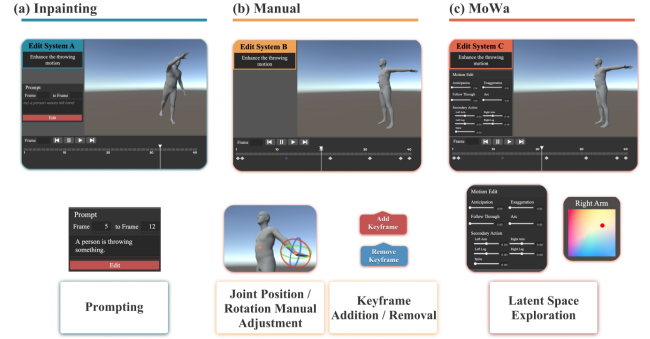


Figure 12: Editing systems tested in T2: (a) The UI of the *Inpainting* system, enabling users to edit motion animations through text prompts; (b) The UI of the *Manual* system, requiring users to manually adjust the rotation and position of avatar joints; (c) The UI of *MoWa*, which allows users to refine motion animations based on animation principles.

the result, using each of the three editing systems in a counterbalanced order. The editing systems were labeled as Edit System A, B, and C, without disclosing their actual names.

Inpainting. The first editing system we tested in T2 is *Inpainting*. The *Inpainting* system uses prompting to edit the given motion. Previous studies have suggested inpainting to edit AI-generated motions [33, 38, 116]. Commercial motion generation online services, such as SayMotion [1], also utilize the inpainting method for editing AI-generated motions. This approach allows users to specify parts of the motion they wish to edit and then re-prompt with the text-to-motion model to regenerate the specified segments. Figure 12-(a) illustrates the UI of *Inpainting* editing system implemented in T2. In the *Inpainting* system, participants can specify a desired range of frames and re-prompt the system to generate motions for the selected range. The prompt is passed to the motion inpainting model [33], which regenerates the specified motion segment.

Manual. The second editing system is *Manual*. The *Manual* editing system emulates a motion editing method commonly found in 3D authoring tools such as Blender, Maya, 3DS Max, and Cinema 4D. Commercial services such as Unity Muse [2] also utilize this approach to edit AI-generated motions. In these systems, keyframes of the motion animations are provided, allowing users to edit the motion by adjusting the poses at each keyframe. Following these systems, users directly manipulate the avatar's joints to modify its position and rotation in the *Manual* condition. Figure 12-(b) shows how the UI of *Manual* system implemented in T2. The participants could select the avatar's joints they wished to adjust and alter their position or rotation as desired. The timeline interface enabled participants to refine motions by adjusting, adding, or removing keyframes until satisfied with the movement.

MoWa. The last editing system is *MoWa*. Figure 12-(c) illustrates the implementation of the *MoWa* editing system and how participants use it to achieve the desired motion outcome in T2. In this editing system, participants can modify the motion by adjusting the scaling factors of the *MoWa* refinement algorithms introduced in

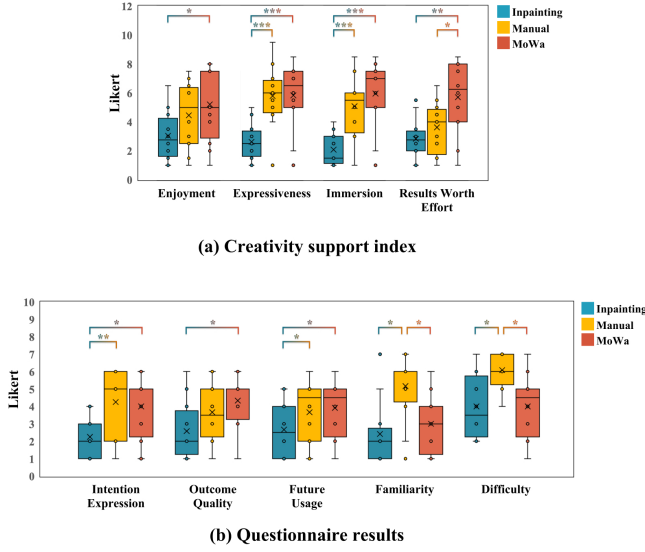


Figure 13: Results from T2: (a) Questionnaire results inspired by the CSI. (b) Additional questionnaire results reported by the designers. Significance levels are indicated as follows: * $p < 0.05$, ** $p < 0.01$, * $p < 0.001$.**

Section 4.4. Adjustments are made through sliders that allow users to increase or decrease the intensity of each scaling factors. For participants requiring more detailed adjustments, the system allows waveform editing by moving points within the rectangular editing interface, allowing participants to make holistic adjustments to poses at each keyframe.

5.3.2 T2: Measures. After editing a motion with each of the *Inpainting*, *Manual*, and *MoWa* systems, participants were asked to evaluate the workflows of each condition using the questionnaires shown in Table 6 – Evaluation. These questions are inspired by previous studies on user experience with AI-driven authoring tools [40, 67, 88] and CSI. After T2, participants completed a comparison questionnaire shown in Table 6 – Comparison questionnaire to rank the editing systems.

5.3.3 T2: Results. We conducted a one-way repeated measures ANOVA (RM ANOVA) to assess the significance between the *Inpainting*, *Manual*, and *MoWa* systems. The sphericity of the data were assessed with Mauchly's test, and we performed post-hoc tests with Bonferroni–holm correction. Significance levels are indicated as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Detailed statistics are presented in Appendix D– Table 8, and the RM ANOVA results are presented in Appendix D– Table 9.

Inpainting: Full Automation, Limited Control. The *Inpainting* editing system received significantly lower user experience ratings compared to both *Manual* and *MoWa*. The user experience results are shown in Figure 13. Specifically, when comparing *Inpainting* to *Manual*, the *Manual* system was rated significantly higher in terms of expressiveness ($p < 0.001^{***}$), immersion ($p < 0.001^{***}$), control ($p < 0.003^{**}$), and future usage ($p < 0.021^{*}$). However,

participants reported that the difficulty of the workflow was significantly lower with *Inpainting* compared to *Manual* ($p < 0.017^{*}$). Comparing *Inpainting* to *MoWa*, participants rated *Inpainting* significantly lower in terms of enjoyment ($p < 0.023^{*}$), expressiveness ($p < 0.001^{***}$), immersion ($p < 0.001^{***}$), results worth effort ($p < 0.007^{**}$), control ($p < 0.011^{*}$), outcome quality ($p < 0.016^{*}$), and future usage ($p < 0.021^{*}$). This suggests that while *Inpainting* facilitates simple editing through text prompts, it was less satisfying compared to *MoWa*.

Manual: High Control, High Effort. In the *Manual* condition, participants reported the highest familiarity with the editing system compared to *Inpainting* ($p < 0.011^{*}$) and *MoWa* ($p < 0.036^{*}$). This familiarity granted participants the greatest control over the motion editing process among the three conditions. However, as shown in Figure 13, no significant difference in the result worth effort metric was found between *Manual* and *Inpainting* ($p < 0.380$), whereas *Manual* scored significantly lower than *MoWa* ($p < 0.037^{*}$). This indicates that in T2, despite the *Manual* condition being the easiest for participants to adapt to, considerable effort was required to achieve a satisfactory outcome. Consequently, the difficulty ratings for *Manual* were significantly higher than those for *Inpainting* ($p < 0.017^{*}$) and *MoWa* ($p < 0.017^{*}$): "Manually adjusting joints for every keyframe requires a lot of effort and is difficult" (P8, P12), "It's hard to achieve natural poses when I have to do it all manually" (P1), and "The burden of designing frame by frame makes it hard to be satisfied with the results" (P3).

MoWa: Balancing Efficiency and Quality in Motion Design. As analyzed through user logs, the computation time of *MoWa* averaged 0.056 seconds ($SD = 0.006$). This computation time includes the process of encoding the generated human avatar motion animation into the latent space using the VAE, performing latent space exploration, and decoding the adjusted motion back into the 3D world space.

The results worth effort metric shows that *MoWa* was rated significantly higher than both *Inpainting* ($p < 0.007^{**}$) and *Manual* ($p < 0.037^{*}$), establishing it as the most efficient method. While *Inpainting* allows designers to generate results with minimal effort, the quality of these results does not meet professional standards. On the other hand, *Manual* provides complete control over motion editing but requires considerable effort. Participants appreciated the efficiency of the *MoWa* method: "It's convenient to adjust poses across multiple frames with a slider" (P4), "It's helpful to change values directly related to design principles" (P4, P5, P10), and "It feels like there's less need to manually adjust poses" (P3).

Among the features related to the animation principles, the participants have noticed the arc, exaggeration, and secondary action to be helpful: "The exaggeration features was the most helpful feature" (P3, P5), "The arc and the exaggeration worked as I expected" (P6), "Adding the secondary action made the motion look more dynamic" (P8, P10). Unlike anticipation and follow-through, which affect only the beginning and end of a motion sequence, arc, exaggeration, and secondary action influence a larger portion of the sequence, making them more noticeable. Participants noted that anticipation and follow-through were more effective when appropriately adjusted in conjunction with other features: "Anticipation and follow-through appeared more effective when adjusted in combination with secondary

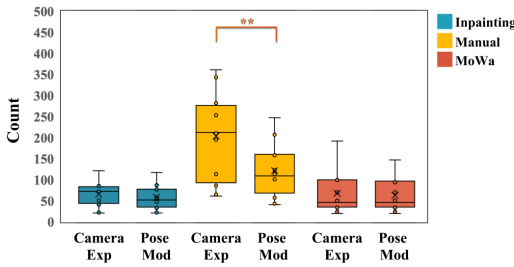


Figure 14: Further observations on operation count in T2. Significance levels are indicated as follows: $*p < 0.05$, $p < 0.01$, $***p < 0.001$.**

action rather than used alone" (P7), and "Follow-through seems to yield better results when adjusted alongside arc" (P1).

Efficiency Analysis on Operation Counts. To analyze how the *Inpainting* and *MoWa* conditions affect efficiency in the design workflow, we examined the operation logs from the participants' tasks. The results are shown in Figure 14 and detailed statistics are shown in Appendix D– Table 9. We categorized the total operations into two groups: operations for modifying the avatar's pose and operations for exploring the world space with the virtual camera. Pose modification operations included using the timeline interface to explore poses, inputting prompts in the *Inpainting* condition, adjusting joint positions and rotations in the *Manual* condition, and modifying parameters or exploring the latent space in the *MoWa* condition. Camera exploration operations included actions such as camera zooming, panning, and dollying. We observed that in the *Manual* system, participants spent significantly more operation counts on camera exploration compared to pose modification ($p = 0.004^{**}$). For *Inpainting* and *MoWa*, the camera exploration was comparable to the pose modification counts. These findings demonstrate that *Inpainting* and *MoWa* methods enhance efficiency by minimizing participants' interactions with the 3D space beyond pose modification.

6 Discussion

6.1 Comparison Between Editing Methods

6.1.1 Inpainting. The *Inpainting* method relied entirely on AI to generate motions, with the AI determining the motion design within the 3D environment. In T1, participants highlighted the convenience of generating reference-worthy motions using text prompts. However, in T2, when editing the synthesized outcome into final design, participants noted that expressing desired motions through text may not fully capture their intentions: "The system was inconvenient because I couldn't directly edit the motion" (P4, P7), "Expressing detailed motion ideas through text doesn't seem effective" (P5, P12), and "Modifying the motion according to my intent was impossible by prompting" (P1). Participants highlighted that the most inconvenient aspect was clearly conveying their intentions in a way that the AI could accurately interpret through text prompts. Consequently, while text prompts are effective for generating initial reference motions, our findings suggest that the *Inpainting* method may not be well-suited for professional designers to edit motion

animations. This is because *Inpainting* does not offer the level of precision required for detailed adjustments, such as specifying exact rotational values for specific joints at intended frames.

6.1.2 Manual. The *Manual* method required participants to control every aspect of the avatar's joint positions and rotations. Participants found the constant interaction to specify joint rotations to be burdensome. Therefore, setting the joint rotation values at each keyframes was considered to be the most noticeable bottleneck of *Manual* method. Due to this burden, although the participants were satisfied with their final design, they found *Manual* to be inefficient: "Manual editing is inconvenient because I have to set everything one by one, but it best expresses what I want" (P3, P8). However, since *Manual* approach also ensures the designer's intention to be accurately reflected, allowing the designers to adjust the motions explicitly in the 3D world space for maximal control is certainly valued. Therefore, while the *Manual* editing system may not efficiently refine motions, it is particularly suited for the final stages of the design process, allowing for direct adjustments to joint angles and timing to achieve the desired pose.

6.1.3 MoWa. *MoWa* provided a middle ground between *Inpainting* and *Manual*. By incorporating priors on human anatomy, *MoWa* offered a partially automated design workflow, ensuring that edits resulted only in physically plausible poses. Compared to the *Inpainting* method, participants reported experiencing greater control during the refining process, making it more suitable for achieving results aligned with their intentions. However, while *MoWa* offered an efficient way to interact with the avatar's pose, its indirect approach to adjusting poses in the world coordinate was less effective for fine-tuning motion details compared to the *Manual*: "I still want to edit the final details by editing the joint rotations" (P6, P9). When finalizing their designs, professional designers required only minor modifications, valuing degrees of freedom over design efficiency. At this stage, the greater flexibility of *Manual*, which allows access to all joint rotation parameters, was preferred over the rapid design space exploration provided by *MoWa*. Therefore, combining *MoWa* for narrowing the design space and refining AI-generated motion animations during the intermediate workflow with the *Manual* editing system for user-directed adjustments in the final stages can address the limitations of both editing systems, resulting in polished outcomes.

6.2 Leveraging Latent Space for 3D Motion Designing

6.2.1 The Effect of Dimensionality Reduction on Editing Expressiveness. The reduction of dimensions simplifies the design process by compressing the design space while retaining the key features of a motion [86]. Previous studies have demonstrated that dividing the design process into manageable subdomains decreases task complexity, making the process more accessible [15]. Similarly, by mapping motion into a low-dimensional latent space, *MoWa* reduced the degrees of freedom, enabling designers to refine motions using a smaller set of sliders. Consequently, participants in T2 appreciated this approach: "Utilizing sliders for editing motion was simple to use." (P10) and "The sliders edited the overall feel of motion, which was efficient." (P3). By focusing on fewer parameters,

designers were able to streamline their workflow and reduce the burden of excessive editing.

6.2.2 Advantages and Disadvantages of Utilizing Latent Space on Editing Poses. *MoWa* offered exploration inside the latent space, which avoided the need for direct exploration in the 3D world space. Navigating the 3D space using a 2D screen projection can create a mismatch between the navigation goals and the desired points due to the ambiguity of 2D projections [37, 75, 79]. Therefore, to overcome this mismatch, participants had to continuously interact with the world space camera when editing with *Manual*, resulting in a high operation count on exploring the 3D space rather than directly manipulating poses. By incorporating latent space, *MoWa* allowed participants to focus more on pose adjustments without the added burden of navigating the 3D environment, leading to a more streamlined and efficient editing process.

However, although the user experience of *MoWa* in T2 was comparable to *Manual*, professionals reported that *MoWa* felt less intuitive. While participants found waveform manipulation intuitive due to its alignment with animation principles, fine adjustments were challenging as they struggled to predict exact outcomes. To address this issue, integrating principal component analysis (PCA) algorithms and visualizing expected outcomes could enhance usability. PCA can help users navigate towards their desired results [21, 28, 58] by translating latent vectors into human-interpretable forms such as text or images [56]. Enhancing interpretability could bridge the gap between the latent space and world space, resulting in a more efficient and intuitive motion design process.

6.3 Adopting AI-driven Authoring Tool into Actual Workflow

Adopting new techniques into the professional workflow can present challenges when integrating them with existing processes [61, 89]. In our user study, participants noted that they were initially unfamiliar with the AI-driven design process, making it difficult to fully utilize the tools in a short time: *"It took some time to understand functionalities of given tool."* (P6). For instance, designers encountered difficulties in identifying effective prompting strategies in T1. While text-to-motion models are generally trained on detailed full sentences describing motions with specific body part movements, such as *"A man throws a ball with right hand quickly"*, designers in T1 often relied on simple and broad keywords like *"Throwing movement"* to generate reference motions. This mismatch highlights that learning effective prompting strategies for AI can act as a barrier for designers. Regarding the usability of integrating AI into workflows, participants noted that instead of relying solely on *Inpainting* or *MoWa*, they preferred combining these tools with their conventional *Manual* methods, which could potentially enhance their workflow: *"I think MoWa would go great with our conventional editing tools."* (P2, P6, P7). Since compatibility with existing methods is a crucial factor in adopting new tools [50, 61, 66], the *Inpainting* and *MoWa* features, which does not interfere with the *Manual* method, can be integrated into existing workflows as additional approaches for efficiently editing motion.

7 Limitations and Future Works

While this study demonstrates the potential of *MoWa* to refine AI-generated motions, several limitations should be addressed to enhance its effectiveness and applicability. First, our study utilized the SMPL format to interface with the outcomes of existing text-to-motion models. Although SMPL simplifies the human shape and pose into numeric parameters, it only represents body poses, excluding details such as facial expressions and finger movements [11]. Future studies should consider extending the system to support the SMPL-X format [70], which also parameterizes facial expressions and finger movements. Second, our approach can be extended to consider interactions with surrounding objects. For example, when exaggerating a motion such as climbing stairs, the edited motion might cause the character's foot to overshoot the step, potentially distorting the intended motion. Future studies could explore the integration of physical simulation based on reinforcement learning to adjust edited motions in response to environmental factors [108–110]. These methods simulate virtual forces applied to an avatar, incorporating environmental effects into the motion. By integrating such simulations, our system could be expanded to maintain context-sensitive accuracy. Third, because professional workflows often require motion animations for non-human avatars, extending our work to a wider range of avatar types should be considered. Previous studies on motion retargeting have explored effective techniques for transferring human motion animations to non-human avatars, facilitating their application across different avatar types [83, 104]. Incorporating these techniques could expand the potential outcomes of our system, as retargeting methods also leverage the SMPL format. Furthermore, while motion synthesis research has predominantly focused on human motion generation, recent studies have proposed initial approaches for synthesizing animal motions using smaller animal motion datasets [107]. Integrating these advancements could further broaden the applicability and usability of our system.

8 Conclusion

This study introduced *MoWa*, an authoring tool designed to enhance AI-generated human avatar motions through latent space exploration. By integrating *MoWa* into professional workflows, we aimed to address the current limitations of AI-generated motions, focusing on improving expressiveness. Our findings demonstrate that *MoWa* streamlines the motion design process, reducing the need for manual adjustments and offering professional designers comparative control over motion outcomes. Through the user study, we observed that waveform manipulation in latent space can resolve the ambiguity issues that arise when navigating 3D space and simplify the design process by mapping complex motions to a low-dimensional space. Future work will explore extending *MoWa*'s capabilities, such as incorporating more detailed motion elements including facial expressions and object interactions, to further support complex motion design tasks.

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A Questionnaires used in Formative Study

Table 4: Questionnaires given in the formative study: The questionnaires aimed to obtain insights on the strengths and weaknesses of AI-generated motions, as well as the refinements required to meet professional standards.

Formative study	Measure	Question	Answer type
Evaluation	Design Level	How well is the motion designed?	Likert (1-7)
	Strength	What are the well-designed aspects of the presented motion?	Open-Ended
	Weakness	What are the shortcomings of the presented motion?	Open-Ended
Post-Questionnaire	Challenges	What challenges arise in designing motion animations?	Open-Ended
	AI Integration	How can AI be integrated into the motion design workflow?	Open-Ended

B Interpolation Functions

- **Vertical Transformation:** Vertical transformation scales the amplitude of the latent vectors within a segment to amplify or attenuate the motion. To ensure continuity at the segment boundaries, the scaling is applied uniformly across the segment. The transformation is expressed as:

$$\mathbf{z}'_j(t) = \mathbf{z}_j(t) + S_V \cdot (\mathbf{z}_j(t) - \mu(\mathbf{z}_j(t))),$$

where $\mathbf{z}_j(t)$ represents the element of latent vector over time, $\mathbf{z}'_j(t)$ is the transformed latent vector, and S_V is the vertical scaling factor. The scaling factor S_V is determined by the height difference between the adjacent extrema of the respective segment.

- **Horizontal Transformation:** Horizontal transformation stretches or compresses the segment along the time axis, altering the duration of the motion. The latent vector is interpolated to account for the time transformation while maintaining continuity at the segment boundaries. The transformation is defined as:

$$\mathbf{z}'_j(t) = \mathbf{z}_j \left(t_{initial} + \frac{(t - t_{initial})}{S_H} \right),$$

where $\mathbf{z}_j(t)$ represents the element of latent vector over time, S_H represents the horizontal scaling factor, and $t_{initial}$ is the start frame of the segment. The scaling factor S_H is determined by the frame difference between the adjacent extrema of the respective segment.

- **Smooth Transition at Segment Boundaries:** To ensure smooth transitions between adjacent segments, a sliding window smoothing technique is applied at the segment boundaries. The smoothed latent vector is calculated as:

$$\mathbf{z}'_j(t) = \frac{1}{2W+1} \sum_{t'=-W}^W \mathbf{z}_j(t+t'),$$

where $\mathbf{z}'_j(t)$ is the smoothed element of the latent vector, W is the window size. This smoothing operation blends the values of adjacent segments, minimizing abrupt changes and ensuring seamless transitions. In our implementation, we set W to be 2.

C Questionnaires used in T1 and T2

Table 5: Questionnaires used in T1: The questionnaire was designed to assess both participants' satisfaction and the efficiency towards the motion generation process.

T1	Observation	Question	Answer Type
Evaluation	Outcome Satisfaction	I am satisfied with the results found through the system.	Likert (1-7)
	Outcome Diversity	The system allowed me to find diverse results.	Likert (1-7)
	Outcome Quality	The quality of the outcomes obtained through this system is high.	Likert (1-7)
	Intention Reflection	The system understood my intentions well.	Likert (1-7)
	Further Usage	I would like to continue using this system in the future.	Likert (1-7)
Comparison	Exploration 1	It was easy for me to explore many different ideas, options, designs, or outcomes.	Likert (1-10)
	Exploration 2	The system was helpful in allowing me to track different ideas, outcomes, or possibilities.	Likert (1-10)
	Preference	Which system do you prefer more?	Binary choice
	Description	Why do you think the system you chose is better?	Open-Ended

D Statistics and Significance Test Results

Table 6: Questionnaires used in T2: : The questionnaire was designed to assess participants' satisfaction with both the generated motion and the process.

T2	Observation	Question	Answer Type
Evaluation	Control	I was able to express my design intentions in the system.	Likert (1-7)
	Outcome Quality	The quality of the outcomes obtained through this system is high.	Likert (1-7)
	Future Usage	I would like to continue using this system in the future.	Likert (1-7)
	Familiarity	I am familiar with this system.	Likert (1-7)
	Difficulty	It was difficult to create the desired outcome.	Likert (1-7)
	Enjoyment 1	I would be happy to use this system on a regular basis.	Likert (1-10)
	Enjoyment 2	I enjoyed using the system or tool.	Likert (1-10)
	Expressiveness 1	I was able to be very creative while doing the activity inside this system or tool.	Likert (1-10)
	Expressiveness 2	The system or tool allowed me to be very expressive.	Likert (1-10)
	Immersion 1	My attention was fully tuned to the activity, and I forgot about the system that I was using.	Likert (1-10)
	Immersion 2	I became so absorbed in the activity that I forgot about the system that I was using.	Likert (1-10)
	Results worth Effort 1	I was satisfied with what I got out of the system.	Likert (1-10)
	Results worth Effort 2	What I was able to produce was worth the effort I had to exert to produce it.	Likert (1-10)
Comparison	Preference	Which system do you prefer more?	Ranking
	Description	Why do you think the system you chose is better?	Open-Ended

Table 7: Significance test results from T1.

Wilcoxon signed-rank			Wilcoxon signed-rank		
Prompt Attempt - Running			Prompt Attempt - Jumping		
Baseline	M = 3.750, SD = 2.491	W = 7.000	Baseline	M = 2.750, SD = 2.301	W = 0.000
MoWa	M = 1.833, SD = 1.030	p = 0.040*	MoWa	M = 1.417, SD = 1.165	p = 0.034*
Prompt Attempt - Throwing			Prompt Attempt - Overall		
Baseline	M = 2.750, SD = 1.913	W = 3.000	Baseline	M = 3.056, SD = 2.150	W = 8.000
MoWa	M = 1.333, SD = 0.651	p = 0.022*	MoWa	M = 1.528, SD = 0.594	p = 0.029*
Exploration Time - Running			Exploration Time - Jumping		
Baseline	M = 154.728, SD = 86.430	W = 19.000	Baseline	M = 107.847, SD = 90.366	W = 13.000
MoWa	M = 101.387, SD = 47.438	p = 0.129	MoWa	M = 72.390, SD = 58.460	p = 0.042*
Exploration Time - Throwing			Exploration Time - Overall		
Baseline	M = 133.177, SD = 72.104	W = 9.000	Baseline	M = 129.139, SD = 70.655	W = 8.000
MoWa	M = 69.936, SD = 48.816	p = 0.016*	MoWa	M = 81.235, SD = 29.499	p = 0.012*
Outcome Satisfaction			Outcome Diversity		
Baseline	M = 2.806, SD = 0.784	W = 74.000	Baseline	M = 2.472, SD = 0.771	W = 72.000
MoWa	M = 4.861, SD = 1.410	p = 0.007**	MoWa	M = 4.139, SD = 1.396	p = 0.001**
Outcome Quality			Intention Reflection		
Baseline	M = 2.333, SD = 0.711	W = 74.000	Baseline	M = 3.028, SD = 0.870	W = 78.000
MoWa	M = 4.167, SD = 1.617	p = 0.007**	MoWa	M = 5.306, SD = 1.029	p = 0.002**
Further Usage					
Baseline	M = 2.333, SD = 0.932	W = 54.000			
MoWa	M = 4.500, SD = 1.982	p = 0.008**			
Exploration					
Baseline	M = 4.000, SD = 1.879	W = 77.000			
MoWa	M = 6.333, SD = 2.490	p < 0.001***			

Table 8: Statistics of each measurements from T2.

Statistics		Statistics	
Operation Count - Camera View		Operation Count - Pose Edit	
<i>Inpaint</i>	$M = 65.667, SD = 28.729$	<i>Inpaint</i>	$M = 58.667, SD = 29.015$
<i>Manual</i>	$M = 203.583, SD = 103.437$	<i>Manual</i>	$M = 121.583, SD = 63.148$
<i>MoWa</i>	$M = 67.583, SD = 49.057$	<i>MoWa</i>	$M = 63.083, SD = 38.888$
Enjoyment		Expressiveness	
<i>Inpaint</i>	$M = 3.042, SD = 1.671$	<i>Inpaint</i>	$M = 2.625, SD = 1.264$
<i>Manual</i>	$M = 4.458, SD = 2.291$	<i>Manual</i>	$M = 6.000, SD = 1.692$
<i>MoWa</i>	$M = 5.208, SD = 2.500$	<i>MoWa</i>	$M = 5.833, SD = 2.319$
Immersion		Results worth effort	
<i>Inpaint</i>	$M = 2.083, SD = 1.062$	<i>Inpaint</i>	$M = 2.833, SD = 1.371$
<i>Manual</i>	$M = 5.292, SD = 1.725$	<i>Manual</i>	$M = 3.583, SD = 1.794$
<i>MoWa</i>	$M = 5.958, SD = 2.350$	<i>MoWa</i>	$M = 5.708, SD = 2.518$
Control		Outcome Quality	
<i>Inpaint</i>	$M = 2.250, SD = 1.138$	<i>Inpaint</i>	$M = 2.583, SD = 1.621$
<i>Manual</i>	$M = 4.417, SD = 1.676$	<i>Manual</i>	$M = 3.833, SD = 1.403$
<i>MoWa</i>	$M = 4.000, SD = 1.809$	<i>MoWa</i>	$M = 4.333, SD = 1.435$
Future Usage		Familiarity	
<i>Inpaint</i>	$M = 2.500, SD = 1.508$	<i>Inpaint</i>	$M = 2.417, SD = 1.832$
<i>Manual</i>	$M = 4.000, SD = 1.595$	<i>Manual</i>	$M = 5.167, SD = 1.899$
<i>MoWa</i>	$M = 3.917, SD = 1.564$	<i>MoWa</i>	$M = 3.000, SD = 1.651$
Difficulty			
<i>Inpaint</i>	$M = 4.000, SD = 2.000$		
<i>Manual</i>	$M = 6.083, SD = 4.000$		
<i>MoWa</i>	$M = 4.000, SD = 1.000$		

Table 9: Significance test results from T2.

RM ANOVA	Post-hoc test		RM ANOVA	Post-hoc test	
Operation Count - Camera View			Operation Count - Pose Edit		
$F(1.341, 14.753) = 3.169$	<i>Inpaint - Manual</i>	$p < 0.001^{***}$	$F(2, 22) = 8.672$	<i>Inpaint - Manual</i>	$p = 0.023^*$
$p = 0.001^{***}$	<i>Inpaint - MoWa</i>	$p = 0.891$	$p = 0.002^{**}$	<i>Inpaint - MoWa</i>	$p = 0.734$
$\chi^2(2) = 6.758, p = 0.034$	<i>Manual - MoWa</i>	$p < 0.001^{***}$	$\chi^2(2) = 4.093, p = 0.129$	<i>Manual - MoWa</i>	$p = 0.011^*$
Enjoyment			Expressiveness		
$F(2, 22) = 4.453$	<i>Inpaint - Manual</i>	$p = 0.136$	$F(2, 22) = 12.436$	<i>Inpaint-Manual</i>	$p < 0.001^{***}$
$p = 0.024^*$	<i>Inpaint - MoWa</i>	$p = 0.023^*$	$p < 0.001^{***}$	<i>Inpaint - MoWa</i>	$p < 0.001^{***}$
$\chi^2(2) = 2.017, p = 0.365$	<i>Manual - MoWa</i>	$p = 0.320$	$\chi^2(2) = 0.640, p = 0.726$	<i>Manual - MoWa</i>	$p = 0.829$
Immersion			Results worth effort		
$F(2, 22) = 14.978$	<i>Inpaint - Manual</i>	$p < 0.001^{***}$	$F(2, 22) = 26.688$	<i>Inpaint - Manual</i>	$p = 0.380$
$p < 0.001^{***}$	<i>Inpaint - MoWa</i>	$p < 0.001^{***}$	$p = 0.007^{**}$	<i>Inpaint - MoWa</i>	$p = 0.007^{**}$
$\chi^2(2) = 0.536, p = 0.765$	<i>Manual - MoWa</i>	$p = 0.388$	$\chi^2(2) = 0.215, p = 0.898$	<i>Manual - MoWa</i>	$p = 0.037^*$
Control			Outcome Quality		
$F(2, 22) = 8.245$	<i>Inpaint - Manual</i>	$p = 0.003^{**}$	$F(2, 22) = 5.047$	<i>Inpaint - Manual</i>	$p = 0.077$
$p = 0.002^{**}$	<i>Inpaint - MoWa</i>	$p = 0.011^*$	$p = 0.016^*$	<i>Inpaint - MoWa</i>	$p = 0.016^*$
$\chi^2(2) = 0.993, p = 0.627$	<i>Manual - MoWa</i>	$p = 0.469$	$\chi^2(2) = 3.745, p = 0.154$	<i>Manual - MoWa</i>	$p = 0.388$
Future Usage			Familiarity		
$F(2, 22) = 5.582$	<i>Inpaint - Manual</i>	$p = 0.021^*$	$F(2, 22) = 5.838$	<i>Inpaint - Manual</i>	$p = 0.011^*$
$p = 0.011^*$	<i>Inpaint - MoWa</i>	$p = 0.021^*$	$p = 0.009^{**}$	<i>Inpaint - MoWa</i>	$p = 0.499$
$\chi^2(2) = 3.112, p = 0.211$	<i>Manual - MoWa</i>	$p = 0.087$	$\chi^2(2) = 3.206, p = 0.201$	<i>Manual - MoWa</i>	$p = 0.036^*$
Difficulty					
$F(2, 22) = 6.233$	<i>Inpaint - Manual</i>	$p = 0.017^*$			
$p = 0.007^*$	<i>Inpaint - MoWa</i>	$p = 1.000$			
$\chi^2(2) = 0.723, p = 0.697$	<i>Manual - MoWa</i>	$p = 0.017^*$			
Further observations					
Inpaint Operation Count	Camera View	$t(11) = 2.217$	Manual Operation Count	Camera View	$W = 66.000$
	Pose Edit	$p = 0.057$		Pose Edit	$p = 0.004^{**}$
MoWa Operation Count	Camera View	$W = 10.000$			
	Pose Edit	$p = 0.100$			